

Table of Contents

Executive Summary	2
1. Verification Campaign - Every Test, Every Result	3
1.1 Method: four independent layers	3
1.2 Static gates: types and lint	3
1.3 The regression battery - and one environment tale	3
1.4 End-to-end: 19/19 in a live browser	5
1.5 Production probes: the deployment is current	6
2. Product Inventory - What Exists Today	6
2.1 The platform at a glance	6
2.2 Learning spine	7
2.3 Exam preparation	7
2.4 Reference library	7
2.5 AI tooling	7
2.6 Test of Control - the v37 flagship	8
3. Health Assessment - Strengths, Risks and Debt	9
3.1 Strengths	9
3.2 Risk register	9
4. Roadmap - What We Can Do Next	10
4.1 Quick wins (days)	10
4.2 Mid-term (next one to two releases)	11
4.3 Strategic horizon (quarter and beyond)	11
4.4 Recommended scope for v38	11
5. Release History and Reproducing the Audit	12
5.1 Version timeline v20 to v37	12
5.2 Reproducing this audit	12

Executive Summary

AuditEdge Academy stands at version 37.0.0, live in production at auditedge-snowy.vercel.app, with the Test of Control section - the newest flagship feature - shipped, verified and deployed. This report answers two questions with evidence rather than impressions: where exactly is the product today, and what should be built next. To answer them, every layer of verification the project owns was executed in a single campaign: static type and lint analysis, the full 21-suite regression battery, a live browser end-to-end walkthrough of the newest feature, and direct probes of the production deployment. The headline is unambiguous: every layer is green. The only defect surfaced in the entire campaign was an environment issue - an empty sandbox database - not a code defect, and it was diagnosed and repaired within minutes.

75,783 Lines of TypeScript in src/	22 Views in the hash router	2,117 Regression checks, all passing	45 Industries in the ToC library
2,125 IFRS revision blocks (41 standards)	2,685 Bank questions in the exam engine	990 Lessons across 41 courses	30 Self-hosted DipIFR exam papers

One operational finding deserves attention before anything else. In a fresh working environment the local SQLite database starts as a 274 KB skeleton, and the regression battery will report fifteen content failures - every one a zero count - until the shipped 4.2 MB content snapshot is restored. The restore is a single idempotent command, but nothing today performs it automatically. Making the test chain self-heal on this condition is the cheapest reliability win available and sits first on the recommendation list.

Three recommendations follow from the full analysis. First, harden the environment: auto-restore the content snapshot when the bank query returns zero, and add retry with backoff to the AI pool, which showed live rate-limiting (HTTP 429) during the model-registry probe. Second, keep deepening the reference library on the proven v36 pattern: IFRS 15 was rewritten to flagship depth (110 blocks) in v36, and IFRS 16, IFRS 9, IAS 12 and IAS 19 are the natural next candidates. Third, grow the Test of Control section from its strong v37 baseline into a fieldwork platform: PDF export of the ICQ working paper, verdict history per entity, and Arabic editions of the industry library.

1. Verification Campaign - Every Test, Every Result

1.1 Method: four independent layers

The campaign was designed as four independent layers, so that a defect missed by one layer would be caught by the next. Static analysis proves the code compiles and conforms to style; the regression battery proves behaviour - engines, scoring, content integrity, generators - against 2,117 assertions; the end-to-end layer drives a real browser through the newest user-facing feature, including a live AI generation; and the production probes confirm that what is deployed on Vercel is actually the version we think it is, serving the assets users reach for. Each layer ran to completion on 5-6 October 2026 inside the development sandbox, against the same commit (0fa90ae) that production serves.

1.2 Static gates: types and lint

Two gates guard every change before it can ship, and both passed clean on the current tree. The TypeScript compiler (tsc --noEmit) reports zero errors across the entire 75,783-line source tree - no unsound narrowing, no broken imports, no silent any-leaks. ESLint over the same tree reports zero violations. Neither tool produced a single warning worth triaging, which is consistent with the project's discipline: every one of the 31 commits shipped through these same gates before this audit ever ran.

1.3 The regression battery - and one environment tale

The regression chain told a cautionary tale worth recording. On its first run the v20 suite reported fifteen failures - but every failing check was a content count of exactly zero: no bank questions, no courses, no lessons, no quizzes. That pattern does not describe broken code; it describes an empty database. The sandbox's local db/custom.db was a 274 KB skeleton, created by a schema push but never seeded. The shipped content snapshot (prisma/auditedge-demo.db.gz, 4.2 MB compressed, 19.59 MB restored) was reloaded with the project's idempotent restore script, which itself verifies the counts it finds: 2,685 bank questions, 41 courses, 990 lessons, 27 quizzes. The chain was then re-run end to end and finished with exit code zero.

Suite	Checks	Outcome
sectors-v15 - audit program library	515	PASS
v23 - ACCA deep past papers	382	PASS
v22 - past-paper bank	366	PASS
v24 - CFA / CMA / CPA skills bank	70	PASS
v37 - Test of Control	75	PASS
v27 - podcast courses and bank	68	PASS
v30 - deep links and exam archive	67	PASS
v26 - flagship question pools	66	PASS

Suite	Checks	Outcome
v32 - IFRS all-41 depth floor	57	PASS
v25 - question generator families	51	PASS
v20 - exam engine, SRS, ISA spine	47	PASS
v34 - DipIFR past papers 2013-2025	48	PASS
v36 - IFRS 15 flagship depth	45	PASS
v31 - IFRS Summaries first wave	44	PASS
v35 - in-app exam viewer	43	PASS
v21 - IFRS 18 and Egypt Arabic parity	32	PASS
v28 - edge TTS voices	31	PASS
models-v15 - AI model registry	31	PASS
engagement-v12 - engagement loop	28	PASS
analyzer-v11 - trial balance analyzer	27	PASS
v29 - engagement smoke	24	PASS
Total - 21 suites	2,117	0 failures

Table 1 - Regression battery results per suite (run of 5-6 October 2026, after content-snapshot restore).

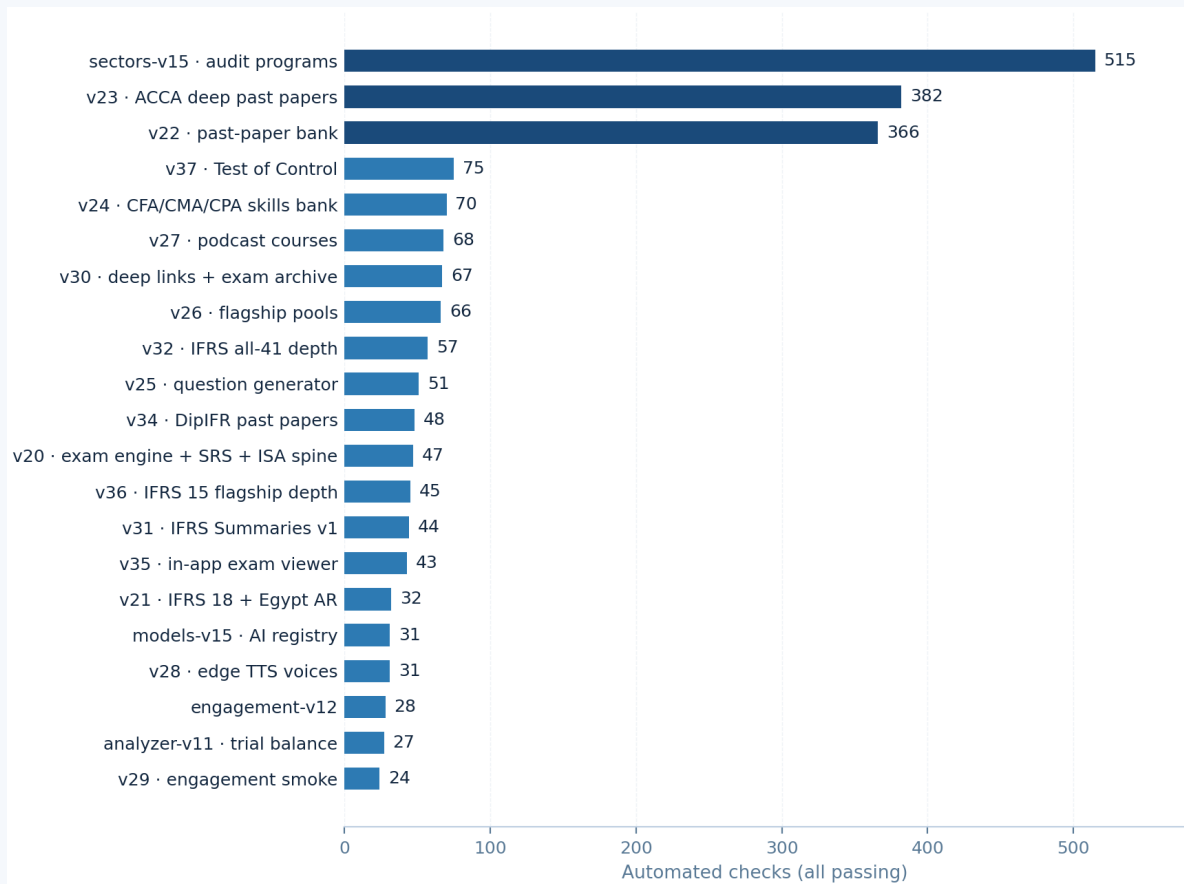


Figure 1 - Regression battery: 2,117 automated checks across 21 suites, zero failures.

The table and Figure 1 show the full distribution. The heaviest suites are the sector-program library (515 checks across the audit programs, their risk mappings and account families) and the two past-paper suites (366 and 382 checks), which validate question format, answer keys, difficulty ladders and bilingual variants item by item. The newest suite - v37, guarding the Test of Control - contributes 75 checks of its own, spanning the industry library's structural integrity (45 industries, 10 sectors, 496 unique questions, no colliding ids) and the verdict engine's scoring rules, including the normalizer that repairs malformed AI output. Summed across all 21 suites the campaign executed 2,117 checks with zero failures.

1.4 End-to-end: 19/19 in a live browser

The end-to-end layer boots the real application - a Next.js dev server, a real browser, real API routes and the live AI pool - and walks the Test of Control feature exactly the way a user would. Nineteen checks passed and none failed. The walkthrough is reproduced here in four groups because it doubles as a feature tour of the newest section.

Navigation and discovery. The sidebar shows the v37 release badge and the new Test of Control entry; the library tab renders industry cards across all ten sectors, from agriculture to awqaf; typing a search term narrows 45 industries to the matching card; the sector chip filter isolates one sector's cards.

The questionnaire. Opening Banking writes its shareable deep link ([#/toc?ind=banking](#)); the full instrument renders 34 questions - the 22-question COSO and ITGC core plus the 12-question banking module - with

three answer options each; the Evaluate button stays disabled until every question is answered.

The verdict engine. Answering everything Yes completes the instrument and returns a Strong control environment at 100 percent with zero critical failures; the domain score bars, the gap list with remediation hints, the corroborating procedures and the one-click markdown export all render on the results page.

The AI generator. The form gates Generate on a non-empty industry description; the API rejects an empty payload with HTTP 400; a real custom request - solar panel cleaning services, an SME with 40 field staff across three cities scheduling in Excel - returned a structurally valid questionnaire of 16 questions and 8 procedures spanning all six COSO domains, which then scored through the same verdict engine.

1.5 Production probes: the deployment is current

The final layer asked the internet what it is actually serving, and confirmed the deployment is genuinely current. The service worker stamps itself auditedge-v37; the built JavaScript chunks served from the CDN contain both the literal string "Test of Control" and the version constant 37.0.0; the new AI route is deployed and enforcing its input guard; the exam archive serves PDFs with the correct MIME type; and the homepage answers in under a second. In short: what the local suites verify is what the world is being served.

Probe	Result
Homepage GET /	HTTP 200 in 0.88 s, 15.9 KB shell
Service worker version stamp	auditedge-v37
Version constant in served chunks	37.0.0 found in deployed bundle
Test of Control UI string in chunks	Present in deployed bundle
POST /api/ai/toc-generate, empty payload	HTTP 400 - input guard live
GET /exams/dipifr/2024-12.pdf	HTTP 200, application/pdf, 966 KB
GET /manifest.webmanifest	HTTP 200

Table 2 - Live production probes against auditedge-snowy.vercel.app.

2. Product Inventory - What Exists Today

2.1 The platform at a glance

AuditEdge Academy is a bilingual (English / Arabic) accounting-education platform for ACCA, IFRS, CPA, CMA and CFA candidates, built as a single Next.js application with a hash-based router serving 22 distinct views, 50 API route files and 54 audit-domain React components - 75,783 lines of TypeScript in all. State persists through Prisma onto SQLite, and offline behaviour is managed by a service worker whose cache stamp is locked in lockstep with the app version, so stale caches self-heal - a lesson learned the hard way in v33 and

automated ever since. The entire interface chrome, on every view, is driven from one i18n dictionary of roughly 1,275 keys carrying full English and Arabic editions, including right-to-left layout switching. Where structure matters, content ships as versioned TypeScript libraries rather than database blobs: the IFRS library (15,493 lines), the audit-program library (5,691 lines) and the new Test of Control library (2,138 lines) are all typed, testable and covered by the regression battery.

2.2 Learning spine

The learning spine carries 41 courses and 990 lessons - the in-house core fully bilingual - plus 27 quizzes wired into a spaced-repetition review engine that schedules recall on an ease-adjusted interval ladder (again-grade repeats in ten minutes, a first good answer returns in a day, an easy answer in three). Progress feeds XP levels, an achievements view and a certificate engine, while a study-plan API spreads preparation across calendar time. The lesson player pairs text with auto-generated voice, and the podcast studio can package any lesson as a voiced bilingual episode.

2.3 Exam preparation

Exam preparation is the deepest vertical. The exam center draws from a 2,685-question bank spanning auditing, financial accounting, Egyptian regulation and ethics, allocated by a blueprint engine - a 45/30/15/10 section weighting - that generates deterministic, non-duplicating sittings of 40 or 60 questions from a single seed. The DipIFR archive, assembled across v34 and v35, self-hosts 30 files (34 MB) covering every sitting from 2013 to 2025 plus companion workbooks, a question index and an English-Arabic glossary - all opened inside the site's own viewer rather than through external links. Two AI routes round out the vertical: an exam generator that assembles papers to a blueprint, and a marker that grades free-text answers. Only the five heaviest files (three workbooks, the BPP kit and the BPP text) remain external downloads, since each exceeds Git's 100 MB object limit.

2.4 Reference library

The reference library is anchored by the IFRS Summaries: all 41 standards in scope, 2,125 revision blocks in total, an average of 51.8 blocks per standard and an enforced floor of 45 - no standard is thin. Blocks are typed (headings, paragraphs, lists, margin notes, formulas, step sequences, classification trees, journal entries, worked examples and exam tips), and journal entries render as T-account tables with code-styled narration strips. The flagship, IFRS 15, was rewritten in v36 to the full depth of the source notes PDF and now stands at 110 blocks with six journal-entry sets. Alongside the summaries sit the audit-program builder - sector programs, risk libraries, methodology, findings, sign-offs, close-out and a prepared-by-client request tracker, guarded by the 515-check sector suite - a KAM drafter for key audit matters, and a trial-balance analyzer that ingests balances and flags the classic error patterns.

2.5 AI tooling

AI tooling runs through one shared model pool with a per-feature API route: a persistent tutor and assistant with conversation history, a podcast studio (edge voices verified across two probes in v28), an industry analyzer, a program tailor, the exam generator and marker pair, and now the ToC generator. Every AI route sits behind

the draft policy - rate-limited and session-gated - and the newest adds a hardened JSON pipeline: replies are forced to JSON, passed through a normalizer that repairs fence-stripping, near-miss domain names, weight drift and duplicate ids, and retried once on structural failure, so the client always receives a runnable questionnaire.

2.6 Test of Control - the v37 flagship

The Test of Control section, shipped in v37, is the newest flagship. It answers one fieldwork question - can this firm's control environment be relied upon? - with three coordinated pieces. The library: 45 industry modules across ten sectors, each carrying 10-13 manager-interview questions with probe hints (what a good answer sounds like), significance weights of 1-3 and critical-control flags, plus 5-8 recommended tests of controls; a universal 22-question core - the COSO 2013 five components plus IT and cyber general controls - opens every questionnaire, so no entity skips the fundamentals. The verdict engine: weighted Yes/No scoring with N/A excluded from the denominator, per-component domain scores, and keystone controls whose failure overrides the aggregate - Strong requires at least 80 percent with zero critical failures, Moderate requires 50 percent with at most two, anything less is Weak. Results render with domain score bars, red-flagged criticals, a gap list with remediation hints, the corroborating procedures mapped to ISA 315/330 strategy, and a one-click markdown working-paper export. The AI generator: any industry not in the library - however niche - plus its case context produces a full bilingual questionnaire, saved locally (capped at twelve) with re-open, delete and regenerate. In total the section carries 496 authored questions, 2,138 lines of library code and 1,579 lines of interface code, guarded by 75 regression checks and the 19-check end-to-end walkthrough described in Section 1.

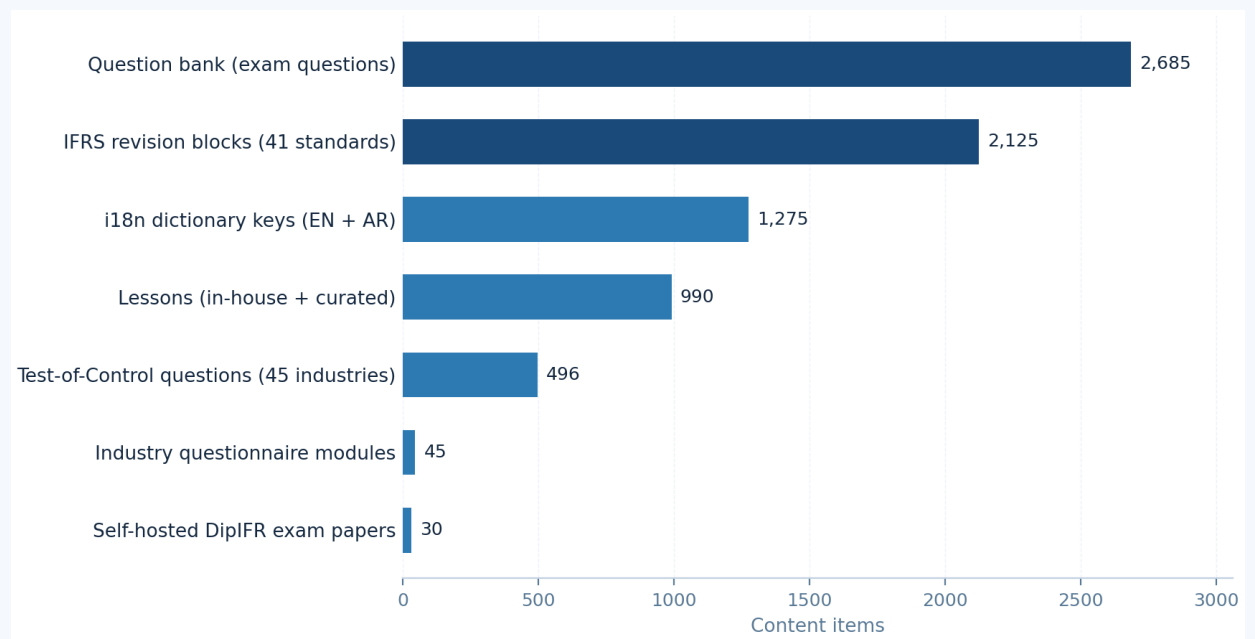


Figure 2 - Content assets by module (live database and content libraries, October 2026).

Asset	Count
Bank questions (exam engine)	2,685

Asset	Count
Courses	41
Lessons	990
Quizzes	27
Test-of-Control questions (45 industries)	496
IFRS revision blocks (41 standards)	2,125
i18n dictionary keys (EN + AR)	~1,275
Self-hosted DipIFR papers + companions	30 files / 34 MB

Table 3 - Content inventory: database and libraries.

3. Health Assessment - Strengths, Risks and Debt

3.1 Strengths

The project's strongest asset is not any single feature but the shipping habit: 31 commits, every version from v20 to v37 released through the same ritual - full battery green, types clean, lint clean, an end-to-end walkthrough, a CHANGELOG entry, then push, with Vercel deploying from main in roughly ninety seconds. The regression suite has grown monotonically with the product - 34 test scripts and 12 end-to-end scripts now live in the repository - which means every feature shipped since v20 is still re-verified on every run. The 2,117-check total is cumulative armor, not a snapshot.

Three structural strengths stand out. Bilingual parity is treated as a first-class invariant rather than an afterthought: one dictionary drives every view, Arabic editions of lessons and quiz variants are asserted by dedicated suites, and right-to-left layout switches with the language. The version number has a single source of truth - package.json - that flows to the build environment, the sidebar badge and the service-worker stamp, with a suite enforcing the lockstep; this is precisely what killed the v33 stale-cache incident class and made staleness visible at a glance. And the exam archive's move in-app (v35) means the heaviest learning assets now ship with the deployment and open in the site's own viewer, with the correct MIME type verified by probe.

3.2 Risk register

Six risks are worth naming. None is a defect today; each is a condition that will eventually cost an afternoon or a user if left alone. They are ordered by expected impact.

Risk	Severity	Impact	Mitigation
AI pool rate-limiting	Medium	A live HTTP 429 was observed during the model-registry probe; the pool is shared capacity, so any AI feature can degrade at peak times.	Retry with exponential backoff, response caching, graceful retry UX.
Manual environment bootstrap	Medium	Fresh clones and sandboxes fail 15 battery checks until the content snapshot is restored by hand.	Auto-restore guard when the bank count reads zero.
Single-user local auth	Medium	One local user, no accounts or roles; blocks team cohorts, shared progress and institutional use.	Multi-user model with roles when collaboration is required.
IFRS depth unevenness	Medium	Only IFRS 15 sits at flagship depth (110 blocks); the other major standards sit at 45-64 blocks against the source notes' full depth.	Continue the v36 pattern: 2-3 flagship rewrites per release.
Heavy exam files external	Low	Five files above Git's 100 MB limit live outside the repository; external links can rot silently.	Periodic link checks; consider LFS or object storage.
Arabic ToC library coverage	Low	Industry questionnaires are authored in English; Arabic output currently relies on the AI generator.	Author Arabic editions for the highest-traffic industries.

Table 4 - Risk register, ordered by expected impact.

Two structural observations complete the assessment. The scripts directory now carries the project's full archaeological record - versioned suites back to v4, seed banks, probes and one-off repairs - which is a strength for verification but deserves a periodic pruning pass of superseded utilities so the suite list stays legible. And the content snapshot doubles as the deployment seed: it is regenerated per release and restored on demand, a mechanism that has proven robust, and one more reason the auto-restore guard recommended in Section 4 matters - it converts the snapshot from a manual ritual into an invariant.

4. Roadmap - What We Can Do Next

4.1 Quick wins (days)

The quick-wins list is deliberately short and deliberately cheap: each item is hours to a day of work, and each either hardens something this audit measured or removes a papercut a real user would feel.

Auto-restore guard. Detect a zero bank count at the start of the test chain and restore the snapshot automatically. Eliminates the only failure mode observed in this entire campaign.

ICQ export to PDF. The markdown working-paper export exists; a print-styled PDF version makes the Test of Control output shareable with engagement partners without a converter.

AI pool retry with backoff. A 429 was observed live; a retry layer with jitter improves every AI feature at once and costs one utility module.

Verdict history per entity. Keep past assessments and show the delta - is the control environment improving or degrading between visits?

4.2 Mid-term (next one to two releases)

The mid-term items are release-sized: one to two releases of focused work each, sequenced to compound.

IFRS flagship deepening. IFRS 16 Leases and IFRS 9 Financial Instruments first (both heavily examined), then IAS 12 Income Taxes and IAS 19 Employee Benefits. The v36 pattern - measure depth in words and journal sets, not block counts - is proven and reusable.

Arabic editions of the ToC library. Translate the industry names, summaries and probe hints for the ten sectors' flagship industries, making the section fully bilingual without AI dependence.

Bank analytics. Surface per-tag miss rates - which standards learners actually fail - and feed them into the study-plan engine so review targets weakness instead of calendar order.

Screening ICQ mode. A 10-question screening questionnaire per industry for first engagements, distinct from the full fieldwork instrument.

4.3 Strategic horizon (quarter and beyond)

The strategic horizon is about who the product serves, not what it contains. None of these belongs in the next release; all of them belong in the decision about the next quarter.

Multi-user teams and roles. Firms running Test of Control across multiple clients; instructors tracking cohorts; shared progress and certificates.

Engagement analytics dashboard. Content-ops visibility: what is studied, where learners stall, which summaries earn re-reads.

Monetization surface. Subscriptions or institutional licensing once accounts and teams exist; the exam archive and the ToC section are the natural paid anchors.

Full offline PWA for the archive. Cache exam papers for offline use, making the app usable on flights and in exam-hall conditions.

4.4 Recommended scope for v38

The cheapest high-value v38 combines the environment guard, the PDF working-paper export, the AI retry layer and one flagship standard - IFRS 16. That is roughly one release-day of plumbing plus one authoring pass, and it hardens everything this audit measured while advancing the reference library. Test of Control depth (verdict history and screening mode) and the second flagship (IFRS 9) then anchor v39, with the Arabic ToC library editions landing alongside whenever authoring capacity allows.

5. Release History and Reproducing the Audit

5.1 Version timeline v20 to v37

Eighteen shipped versions in the tracked window, each carrying its own regression suite forward. The table reads as a product biography: an exam engine first, content depth second, distribution and reliability third, and now fieldwork tooling.

Version	Highlight
v20	Exam engine: blueprint, SRS, ISA spine, Arabic depth pass
v21	IFRS 18 and Egypt regulation Arabic parity
v22	Past-paper question bank (366-check suite)
v23	ACCA deep past papers
v24	CFA / CMA / CPA skills bank
v25	Template-driven question generator families
v26	Flagship pools and generated podcast courses
v27	Podcast studio and bilingual episodes
v28	Edge TTS voices verified and wired
v29	Engagement tuning pass
v30	Shareable deep links and IFRS depth wave one
v31	IFRS Summaries beyond 750 blocks
v32	All 41 standards lifted past 45 blocks
v33	Service-worker self-heal for stale caches
v34	The real DipIFR past papers 2013-2025
v35	Sameh Zidan exams open inside the website
v36	IFRS 15 rewritten to the notes-PDF depth (110 blocks)
v37	Test of Control: 45 industries, verdict engine, AI generator

Table 5 - Release history, v20 through v37 (31 commits on main).

5.2 Reproducing this audit

Any future audit can be reproduced from a clean clone with the commands below, in order, in under ten minutes of wall-clock time. The restore step is idempotent and safe to re-run; the battery gates on it, which is exactly why the auto-restore guard is recommended.

```
# 1. restore the content database (idempotent)
bun scripts/restore-local-db.ts

# 2. static gates
bunx tsc --noEmit && bunx eslint .

# 3. full regression battery - 21 suites, 2,117 checks
bun run test

# 4. end-to-end walkthrough of the newest feature (boots the app)
bash scripts/e2e-v37.sh
```

The campaign's conclusion can be stated in one sentence: version 37 is the healthiest state the product has ever been verified in, with every layer of its own test architecture green and the deployment proven current - which makes this exactly the right moment to spend capacity on depth (flagship standards, Arabic editions) and reach (PDF working papers, verdict history) rather than on repairs.